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The Missing Phase: Why Supply Chain Disruption Response Remains Unoptimised

An evidence-based analysis of the gap in supply chain risk management — and the case for prescriptive optimisation in the respond and recover phases of the disruption lifecycle

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Executive Summary

Supply chain risk management has a blind spot that costs enterprises an estimated $43–47 million per year. The industry has built robust systems for detecting disruptions and quantifying their impact — but the critical next step, determining what to do about it in an optimal, defensible, and time-sensitive way, remains stubbornly manual, heuristic-driven, and slow.

Current SCRM platforms (Resilinc, Interos, Everstream Analytics) excel at telling companies what happened and who is affected. Planning tools (Kinaxis, SAP IBP, o9 Solutions) provide scenario modelling for planned cycles. Network design optimisers help companies design resilient supply chains in advance. None closes the **response optimisation gap**: the real-time, in-disruption decision problem of finding the mathematically optimal mitigation strategy across the full recovery arc.

Five findings anchor the case for prescriptive response optimisation:

FINDING 01

Fewer than 7% of companies introduce genuinely new countermeasures during disruptions (McKinsey, 2025). The other 93% apply the same inventory–dual source–regionalise playbook regardless of disruption type.

FINDING 02

83% of supply chains cannot respond to disruptions within 24 hours. Average response time is five days; average recovery is three to five months.

FINDING 03

A disruption lasting over 30 days costs 30–50% of annual EBITDA — ten times more than one contained within a month. Every day of delayed optimal response amplifies damage nonlinearly.

FINDING 04

The "respond" phase is the least automated phase of the SCRM lifecycle. Academic literature (2025) confirms that mitigation and handling are "significantly harder to digitise" than detection and assessment.

FINDING 05

GenAI alone cannot solve this problem. Its probabilistic outputs, inability to enforce hard constraints, and absence of a formal objective function make it unsuitable as a standalone response tool. The combination of GenAI (for natural language and explainability) with mathematical optimisation solvers (for feasibility and optimality guarantees) is architecturally superior to either alone.

The SCRM Lifecycle Gap

The SCRM lifecycle runs through four phases: Detect (event scouring, alerts), Notify (alert and impact analysis), Respond (options evaluation, mitigation strategy, optimal solution), and Recover (execution and tracking).

Current SCRM incumbents — Resilinc, Interos, Everstream — operate primarily in the Detect and Notify phases. This is confirmed by their own product trajectories: Resilinc's mid-2025 "Agentic AI" launch, explicitly moving toward alternative sourcing recommendations, is an acknowledgment that prior capabilities did not include response optimisation.

The critical gap sits in the Respond phase: finding the optimal response, not just any response. ECC operates across the full SCRM lifecycle — through its alert intelligence partnership it ingests and processes disruption signals at near real-time, and its primary differentiation is in the respond-and-recover phase, where it applies mathematical optimisation to generate the full solution space, co-optimises the disrupted-to-recovered temporal window, and explains every recommendation in terms a planner can defend to leadership.

SCRM Tools Stop Where the Hardest Decisions Begin

The supply chain risk management market — approaching $4.5 billion in 2025 and projected to reach $7–9 billion by 2032 — is dominated by platforms that monitor, detect, and alert. A systematic review of 171 academic papers on SCRM automation (Electronic Markets, November 2025) found that digitalisation "is not uniform across phases" of the SCRM lifecycle. Mitigation and handling "involve complex decision-making and action, making them harder to digitise" — confirmed by a Taylor & Francis (2024) review finding that prescriptive analytics for supply chain resilience is "significantly underexplored compared to its descriptive and predictive counterparts."

The major SCRM vendors confirm this gap through their own product trajectories. Resilinc, the Gartner Magic Quadrant Leader (April 2025), built its reputation on multi-tier mapping and event monitoring. Its mid-2025 Agentic AI launch is an acknowledgment that prior capabilities did not include response optimisation. Customer reviews corroborate: "While the platform excels in risk identification, its mitigation recommendations could be more robust and actionable." Interos describes prescriptive analytics for supply chain orchestration as a roadmap aspiration, not a delivered capability. Everstream Analytics is a risk intelligence overlay — action plans are human-created based on its intelligence, not auto-generated.

The workflow that fills the response gap today is strikingly manual. When a disruption is detected, a cross-functional war room convenes — procurement, supply chain planning, logistics, quality, finance, and executive leadership. McKinsey's 2024 Global Supply Chain Leader Survey found that the average company takes two weeks to plan and execute a response. Accenture found that 57% of companies take more than a week just to be alerted, with average recovery stretching to three to five months.

Planning Tools Were Built for Steady-State, Not Crisis

Kinaxis, SAP IBP, o9 Solutions, Blue Yonder, and Anaplan do not fill the response optimisation gap despite increasingly aggressive marketing. The architectural limitations are structural, not incremental.

Kinaxis RapidResponse saw a 2× surge in usage during COVID-19 as customers actively replanned — but it remains a planning and modelling tool that recommends actions without executing supply chain changes. SAP IBP's Supply Chain Control Tower offers pre-defined response playbooks, but Gartner's 2024 Critical Capabilities report notes that multienterprise and financial impact analysis capabilities are the least-used features. Blue Yonder has the deepest optimisation heritage but achieving instantaneous orchestration requires integration depth that most implementations lack.

Five structural gaps limit all planning tools in disruption contexts:

- **Batch vs. continuous processing.** Most were designed for nightly or weekly planning runs, with data refresh latencies incompatible with in-crisis decision windows.
- **The planning-execution gap.** Generating an optimal plan and actually updating purchase orders, production schedules, and logistics routings are different operations. Planning tools stop at the former.
- **Time fence conflicts.** Planning tools respect frozen planning periods architecturally — disruption response inherently requires violating them.
- **Model assumption failure.** Planning models assume structural constants that break simultaneously during major disruptions. When a port is disrupted, a supplier is down, and demand shifts simultaneously, the model's encoded assumptions become invalid.
- **The optimisation granularity-speed tradeoff.** Full mathematical optimisation of a complex supply network is computationally expensive and incompatible with real-time response, while fast heuristic approximations sacrifice solution quality.

A Two-Phase Decomposition Architecture

One approach to resolving the granularity-speed tradeoff is a two-phase decomposition architecture grounded in established operations research methodology — spatial and geographic decomposition underlies canonical OR methods including Benders decomposition, Lagrangian relaxation, and column generation.

Phase 1 — Local zoning. Each affected plant or facility is treated as an independently parameterised optimisation entity with its own supplier relationships, inventory positions, customer commitments, production constraints, and disruption exposure. Solving locally delivers actionable plans within the first hour of a disruption — before a war room has even fully assembled.

Phase 2 — Network-level expansion. In parallel, the global network problem is solved, capturing cross-plant interactions — particularly shared resource constraints such as two plants drawing from the same constrained alternate supplier. Local plans are updated as the global solution converges.

Teams Default to the Same Playbook Every Time

McKinsey's 2025 survey stated it explicitly: "Most countermeasures follow the same playbook that companies have applied when faced with other large-scale disruptions." For tariff responses: 45% increased inventories, 39% pursued dual sourcing, 33% developed nearshoring plans — the same three levers applied consistently since 2020. Fewer than 7% introduced genuinely novel countermeasures.

The behavioural economics literature explains why. Doyle, Brito, and Dawson (2021) studied frontline supply chain employees and found consistent anchoring effects and overconfidence, with employees resorting to "strategies of 'muddling through'" when lacking time to "access, assess and evaluate the full range of options." Roehrich, Grosvold, and Hoejmose (2014) documented bounded rationality — companies "do what they can" balancing implementation costs against perceived exposure probability.

Why GenAI Alone Does Not Solve the Playbook Problem

GenAI tools do not change this pattern. The reason is structural. GenAI alone has three fundamental limitations in high-stakes operational decisions during a disruption:

- **Probabilistic, not deterministic.** GenAI generates plausible outputs, not provably optimal ones. In a disruption scenario with hard constraints — supplier lead times, container capacities, regulatory compliance, contractual minimums — "plausible" is not sufficient.
- **Cannot enforce hard constraints reliably.** A properly formulated MILP is constitutionally incapable of violating a constraint. GenAI, told that a supplier has a capacity of 500 units, will sometimes generate a plan that exceeds that capacity — particularly when handling multiple interacting constraints simultaneously.
- **No objective function.** GenAI produces recommendations without proving they minimise cost, maximise service level, or optimise any quantifiable goal across the full decision space.

The result: teams using GenAI Q&A tools still default to familiar options — they ask about the alternatives they already know, and the tool obliges. The option generation problem remains unsolved. As the 4I framework paper (Simchi-Levi et al., SSRN 5511218) states: "GenAI lowers barriers to modelling and interpretation, while mathematical optimisation reliably enforces business goals, rules, and hard constraints."

Every Day of Delayed Response Compounds the Damage Nonlinearly

McKinsey estimates that over a decade, supply chain disruptions cost companies approximately 45% of one year's profits. A disruption lasting 30 days or less costs 3–5% of EBITDA; a long-term production disruption costs 30–50% of annual EBITDA — a tenfold increase. An IDC/Kinaxis survey found the average disruption response time is five days, with only 17% of companies able to respond within 24 hours. Accenture estimates disruptions caused companies to miss 7.4–11.0% of revenue growth — $1.6 trillion in missed annual revenue globally.

The cases where optimisation has dramatically compressed response times:

- Microsoft's OptiGuide reduced planner question-response time from 2.5 days to near real-time, cutting fulfillment team time by 23%.
- IBM's cognitive supply chain reduced disruption mitigation per part number from 4–6 hours to seconds, saving $160 million in reduced inventory costs.
- Toyota compressed earthquake recovery from six months (2011) to two weeks (2016) through optimisation and resilience investments.

The Case for Prescriptive Response Optimisation

The financial pain framing is the right starting point: a supply chain disruption lasting over 30 days costs 30–50% of annual EBITDA — ten times more than one contained within a month. The difference is almost entirely in the quality and speed of the response decision. SCRM tools tell you what happened and who is affected. What they do not do is tell you the optimal response — across all options, under all constraints, in time to matter.

Current war-room processes — spreadsheets, conference calls, exhausted experts applying the same three levers — leave significant value on the table. McKinsey found fewer than 7% of companies introduce genuinely new countermeasures during disruptions. The rest do not know what they are missing. A system that generates the full solution space, including options the team would not think to ask about, and ranks them by cost, risk, and delivery impact simultaneously, closes this gap.

The biggest barrier to using optimisation in a crisis is not the mathematics — it is trust. Planners will not act on a recommendation they cannot explain to their leadership. This is why explainability is essential to any response optimisation system: every recommendation must come with a plain-language rationale tied to the actual model logic — which constraints drove the decision, what the cost trade-offs are, and why alternative options were ranked lower.

Current GenAI Systems Answer Questions Asked, Not Questions That Should Be Asked

Current GenAI and optimisation systems are conversational wrappers around solvers — valuable, but constrained to answering questions the planner already knows to ask. The most valuable insight is often an option the planner never considered.

A query-answering system responds to: "What if supplier A goes offline?" → re-solves → returns updated plan. A genuine prescriptive analytics system proactively generates the full solution space — including options the planner never considered — and presents them as a ranked, comparable set.

Capability	Industry Maturity
NL → MILP formulation	Medium-high (research systems available)
What-if Q&A on existing models	High (production deployments exist)
Interpretable optimisation explanations	Medium
Continuous re-optimisation on disruption	Medium-low (controlled settings only)
Autonomous option generation	Very low — no production system
Pareto frontier of response strategies	Very low (OR theory only)
Temporal co-optimisation across recovery arc	None commercially

Where the Current Market Leaves a Gap

No formal analyst category yet exists for "prescriptive optimisation in supply chain disruption response." No Gartner Magic Quadrant, no Forrester Wave. The market currently splits into three non-overlapping categories: SCRM tools (monitoring and visibility — urgent but descriptive); supply chain planning platforms (scenario modelling and execution — prescriptive but designed for planned cycles); and supply chain network design platforms (mathematical optimisation — rigorous but strategic and slow). The combination of operational urgency with prescriptive optimisation has no established incumbent.

The most relevant pure-play optimisation competitor is Lyric.tech (ChainBrain, Inc.), which has raised $67 million and serves 25+ enterprise customers. Its platform combines mathematical optimisation, prediction, and simulation in a composable architecture. The critical distinction: Lyric is a strategic network design and planning optimisation platform. Its customers use it to answer planned-cycle questions made weeks or months in advance, with full data. The unplanned, time-critical, operational disruption response problem — when the designed network has failed and a decision must be made in hours, under uncertainty — is a fundamentally different problem.

Digital Twins, Autonomous Systems, and Trust

Gartner projects that by 2030, half of all supply chain management solutions will employ agentic AI to autonomously execute decisions. Companies using agentic AI outperform competitors by 2.2× in operational resilience. The emerging paradigm shifts from human-in-the-loop to human-on-the-loop — AI executes routine decisions within predefined boundaries; humans oversee and intervene for exceptions.

Algorithm aversion is the binding constraint on adoption. Research examining safety stock decisions during disruptions found that algorithm aversion increases as disruptive events unfold — precisely when algorithmic recommendations are most valuable. The more experienced the decision-maker, the less they trust algorithms for consequential supply chain decisions under crisis conditions. This paradox means that the most sophisticated optimisation will face the strongest resistance from the most experienced practitioners during the highest-stakes moments.

Trust-building requires transparency, explainability, gradual autonomy expansion, and demonstrated learning capability — not just technical superiority. The explainability layer — generating plain-language rationale tied directly to solver artefacts and constraint provenance — is the mechanism that makes optimisation recommendations actionable under crisis conditions, when planners' instinct to override the algorithm is at its strongest.

Optimisation Quality Depends on Visibility and Regulatory Compliance

Two factors fundamentally bound the quality of any disruption response optimisation: how many options the system knows about, and which options are legally permissible.

On visibility: 45% of companies have no supplier visibility beyond Tier 1. A MedTech company with 29 Tier-1 suppliers discovered 212 at Tier 2, 1,766 at Tier 3, and 13,876 at Tier 4 through multi-tier mapping. Optimisation engines operating only on Tier 1 data explore a tiny fraction of the solution space.

On regulation: hard constraints are multiplying. UFLPA enforcement saw 7,325 shipments stopped for review in FY2025 — a 51% year-over-year increase — with noncompliance costing upwards of $810,000 per detention case. The EU CBAM is expanding to chemicals, glass, and paper. The EU CSDDD mandates human rights due diligence throughout value chains. When a preferred supplier is disrupted, switching to a lower-cost alternative may be blocked by UFLPA exposure or CBAM carbon intensity thresholds. Any optimisation that does not incorporate these constraints will generate legally infeasible recommendations. Regulation-aware optimisation for disruption response is a defensible moat for solutions that embed these constraints natively.

Strategic Implications

The evidence assembled in this report points to several conclusions relevant to supply chain risk strategy and technology selection.

The gap is real, documented, and financially significant. Academic literature, analyst reports, vendor roadmaps, practitioner surveys, and customer reviews converge on the same finding: the respond-and-recover phase of the SCRM lifecycle is underserved by formal optimisation. Companies lose $43–47 million annually to disruptions (Interos), take two weeks to respond (McKinsey), and apply the same three levers 97% of the time regardless of disruption type.

The market is moving, but the respond phase remains unclaimed. Resilinc's Agentic AI launch, SAP's autonomous orchestration vision, and the 4I framework all signal that the industry is directionally aware of this gap. No platform has yet closed it. Organisations evaluating technology in this space should scrutinise whether response optimisation capabilities are live and deployed, or roadmap aspirations.

The highest-value capability is option generation, not question answering. The current GenAI and optimisation paradigm is necessary but insufficient. Systems that proactively surface the full set of response strategies — including options practitioners would not think to ask for — represent a meaningfully different capability from those that answer individual queries well. This distinction is worth evaluating explicitly when assessing platforms.

Three factors will determine adoption. First, trust and explainability: algorithm aversion is strongest during high-stakes disruptions, so transparent reasoning and gradual autonomy are prerequisites. Second, data integration: optimisation quality depends on multi-tier visibility and real-time data feeds that most enterprises lack. Third, regulatory awareness: the expanding web of UFLPA, CBAM, CSDDD, and similar regulations creates hard constraints that any feasible optimisation must incorporate.

> The supply chain industry has spent a decade perfecting the ability to see disruptions coming. The next decade belongs to those who can determine what to do about them — not with the same playbook every time, but with mathematically rigorous, contextually aware, regulation-compliant solutions generated faster than a war room can convene.

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